DevOps Deployment & Operations Guide

Service: privacy-detection-service | **Repository:** HekmaAI/privacy-detection-service | **DB:** Existing App DB (hekmaai)

1. Executive Summary & Production Topology

The **privacy-detection-service** provides high-speed PII token classification (sub-40ms RoBERTa) and swappable Clinical PHI detection (OpenAI gpt-4o-mini) storing encrypted surrogate mappings directly in the application database (hekmaai).

2. Multi-Branch & Multi-Environment Matrix

Git Branch	Environment	Custom Domain Endpoint	Database	Service / Deployment Name
develop / dev / Pii	development	https://dev-pii-phi.hekma.ai	hekmaai	privacy-detection-service-dev
demo	demo	https://demo-pii-phi.hekma.ai	hekma_demo	privacy-detection-service-demo
staging	staging	https://staging-pii-phi.hekma.ai	hekma_staging	privacy-detection-service-staging
main	production	https://pii-phi.hekma.ai	hekma_prod / hekmaai	privacy-detection-service-production

3. Container & Compute Specifications

• **Compute: 2 CPU (2000m)** and **4 GB RAM (4096 MB)**.
• **Process Manager:** Gunicorn with 2 Uvicorn async workers on container port 8080.
• **Security:** Runs as non-root user appuser (UID 1001) with read-only root filesystem.
• **Warmup:** RoBERTa model weights load during lifespan startup for instant sub-40ms inference.

4. Environment Variables & Secrets Reference

Variable Name	Source	Purpose / Sample Value
DATABASE_URL	Kubernetes Secret / Vault	postgresql+asyncpg://admin:SECRET@host:5432/hekmaai
OPENAI_API_KEY	Kubernetes Secret / Vault	sk-proj-... (Used strictly for clinical notes)
CUSTOM_DOMAIN	ConfigMap / Env	https://pii-phi.hekma.ai (Portless clean URL)
STORAGE_BACKEND	ConfigMap / Env	postgresql (or json for zero-database test mode)

5. Database Migration Runbook

Creates 4 isolated tables directly in hekmaai: token_mappings, policies, audit_logs, and organizations.

```
# Automated Multi-Environment CLI Migration:
./scripts/migrate_database.sh dev
./scripts/migrate_database.sh demo
PROD_DATABASE_URL="postgresql+asyncpg://admin:SECRET@host:5432/hekma_prod" ./scripts/migrate_database.sh prod
```

6. Observability Probes & Rollback

• **Liveness Probe:** GET /health | **Readiness Probe:** GET /ready | **Metrics:** GET /metrics
• **Rollback:** kubectl rollout undo deployment/privacy-detection-service -n privacy-detection